The Project Management Course: Beginner to PROject Manager

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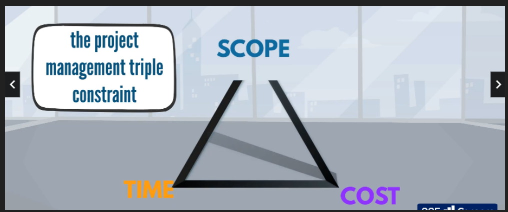
Contents

Chapter 1

Welcome to the course! Introduction to projects and project management

1.1 What is a project?

- A project is:
  - A temporary endeavor undertaken to create a unique product, service or result. PMI definition.
  - It has a determined start and end date.
  - It is not a business process that is ongoing but rather have a specific expiring date.
  - It is time bound and has a goal.
  - The goal can be a result, service or produt.
  - The output + the work employed to create it is called a scope.
- Elements of a project:
  - Time (start and end date) + goal (product, service or result) + cost.

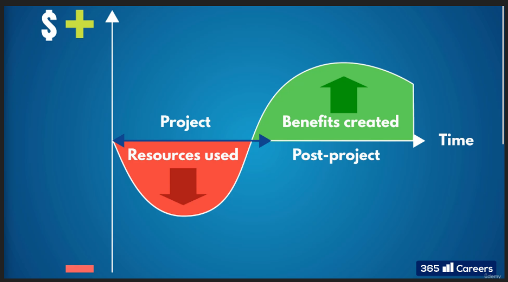


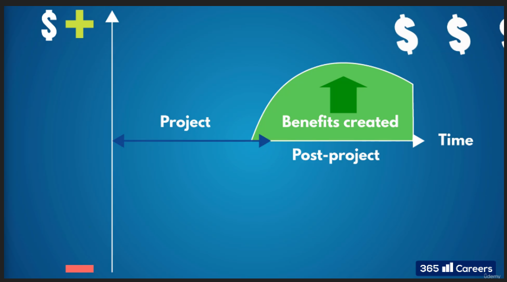
  - These three elements form what is known as a project management triangle where any change in one of the dimentions results in the changing of all the other dimensions, there is a tradeoff between these dimensions, when you want less time this might affect cost, etcetera.
- Ending definition:
  - A project is a temporary initiative that is agreed, planned, and executed to achive a specific goal.
  - Projects are complex initiatives.

1.2 Why do companies execute projects?

- All companies have goals. Some goals are for daily work goals and other goals are strategic.

- To achieve the strategic goals a company cannot lean on the same old strategies but rather create something new.
- In the implementation of any strategic goal you need to achieve in a company, you must do it usign a project. This is why companies execute projects.
- In order to achieve the project as an output you need to put in the following inputs:
  - Resources,
  - Financial Resources,
  - Organizational resources, and
  - Time.
- In highly innovative indutries failing to prioritize getting the necessary resources in the right time might put the company at a significant disadvantage.
- Correct project management derives the following benefits:
  - Financial, operational, customer related, non-profit.
- Let's see the life cycle of a project:





1.3 What creates the demand for a project? Prioritizing and selecting a project.

- A project must relate to a business strategy. So what creates the demand for a project in the first place.
- The following:
  - Market need, to keep up or have advantage over the competition.
  - Business need.
  - Technological advancement. i.e. for example: automation.
  - Customer request. i.e. satisfying client's needs.
  - Legal requirements. For example: social media.
  - Social needs. For example: building a bridge.
  - Ecological impact considerations. For example: improving cars
- Once a specific goal arises, the PROJECT will be the instrument to achieve that goal.

1.3.1 Priritizing and project selection

- The needs most likely will never arive one by one, so how do you choose which ones to do and in what order?
- This is when prioritizing comes in.
- The process is the following:
  - First the project owner will propose the project via a project proposal to management to make it compete for the resources of the organization.
  - Management will decide which project will be discarded, prioritized, or posponed for a later date.
  - Prioritizing urgency: for example if some law requires some inmediate project or else they wont be able to operate, management will prioritize it because it is urgent.
  - This is known as project selection.
- The project selection process selects the order in which each project will be done, and will schedule the order in dates. Which projects will be concurrent etcetera. Management will make what is called a project portfolio, this is a document that contains the details on which projects will be done and in what order, and which project will be done at the same time (concurrently).
- Project Portfolio Management: the process of creating and managing the project portfolio in such a way that it uses the organization's resources in an optimal way.
- Projects are investments and need to be done in as efficient a manner as possible. Projects consume a lot of resources, finances, time, effort, attention, care, etcetera. If they are not aligned and optimized it may result in the organization's bankrupcy or at best to disapointed stakeholders.

1.4 What is a project manager? What is their commitment to a project?

- The project manager is the one responsable for the project, the PM is the CEO of the project.
- The PM person needs to work between these constraints (budget, time, performance) in order to meet the goal and add value to the business.

- This is a big responsability. The PM must therefore choose realistic projects, taking into account all costs, all time constraints. Otherwise they will not be able to delive and will be in problems.
- It is a project manager's duty to do the following to a project:

- Asses, Discuss, Negotiatite, Adjust: demonstrate why the project will be ready in the proposed date, why it will cost the proposed amount. Otherwise refuse to take a project on.

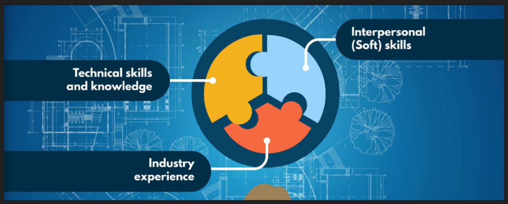
- What does it mean to be accountable?
  - The project manager has the ultimate responsability, they can't blame the failure of a project on any one else. 'The buck stops with them'.
  - Project manager must commit and deliver.

- The PM responsability is not only responsable for themselves but also for the actions of their team members. A PM must be able to have full control, and take inmediate action upon a problem arising.

1.5 What are the skills and the knowledge a PM must have?

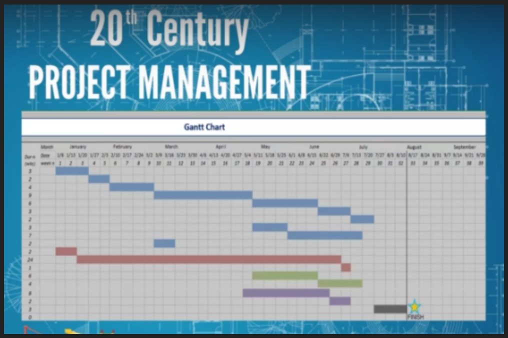
- The PM is the face of the project.
- They are the point of contact of the team.
- Must mediate between team members.
- Successfully relate with team members with all levels of seniority.
- They are accounable to deliver the expectations.
- They must therefore have:
  - Skills, knowledege, attitude, and experience.
- The three major qualities of PM:
  1. Technical skills and knowledge.
     - PM triple constraint.
     - Theoretical framework.
     - PM tools and documents.
     - Personal organization (ability to organize you own tasks).
     - Task management (ability to fulfill tasks, prioritize and manage tasks).
     - Critical thinking (tell which information is useful and which is not).
     - Efficient use of software tools for PM.
  2. Interpersonal soft skills:
     - Each project is unique and must adapt to a new set of persons each time.
     - They need communication skills (90% of their time will be in communications).
     - Team management, being able to asses performance and output of the team, the PM needs to be able to articulate the project's needs and requirements, identify impediments and keep the team motivated.
     - Leadership, ability to influence people in the desired direction, capacity to mediate.
     - Social skills, ability to work with a lot of people.
     - Advice: never talk exclusively about the project, talk also about personal lives and hobbies, otherwise team members will associate you only to office work and you will not be able to influence and motivate them personally. Also be humorous, it relaxes people.
  3. Industry experience:
     - Does the PM needs to be an expert in the industry in which their project are taking place? it depends, knowledge always helps.

  - Being an expert in a field does not imply success in managing the project related to that field. It helps but in order to manage project effectively you need first and formost... project management skills. Industry expertise is a bonus most of the time.



1.6 Projects: a history lesson

- Project management has been going on for thousands of years.
- An example includes the pyramids of ancient egypt. This construction was massive endeavor, involving materials and a workforce. The success of this most likely depended on good project management.
- Another example is when Columbus sailed to America, he did it with a budget of three ships and some ship crew and his proposal was rejected multiple times by various kings. Eventually they found a sponsor of this projects, the king and queen of Spain.
- Another example, 19th century and the industrial revolution, in which project management became more and more popular since more projects were going on.
- Another example, 20th century project management, the Gantt chart was created by Herny Gantt, it was used in WW1 to track and manage navy ship creation.



- In the late 1950s the *critical path method* this is a methodology allows to calculate the fastest way to completing a project, finding links and connections between the activities involved.
- In the 1960s the Project management profession was established. The international project management association and project management institute established standards, best practices, tools and guidelines.
- The 21st century project management scheme is exceptional because the projects are incredibly complex and have many many variables and factors, as well as risks that must be all individually considered. A strong PM is a very important asset in all organizations.

1.7 Project management terminology

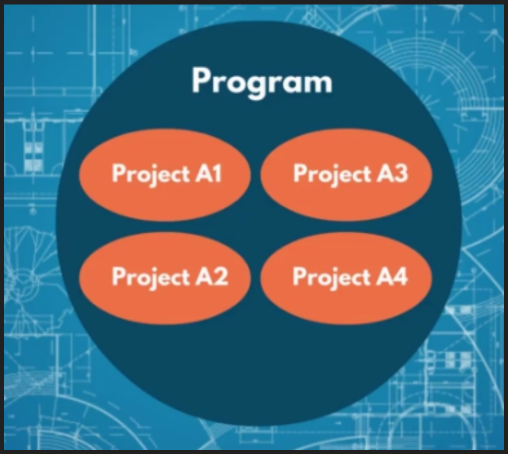
- Project management office (PMO): department reposponsable for managing coordinating and consulting project related work. The PMO will vary in size and structure from company to company.

- Consulting companies will have very large PMO unit. While some others might not even have a PMO unit.

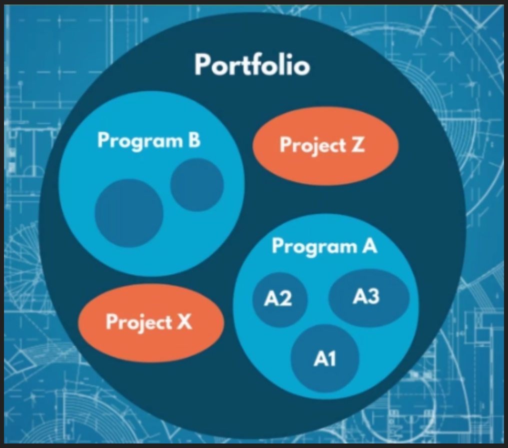
- Project team: experts responsable for the execution of the work. Anyone directly working on the project.
  - Example: developers, supervisors, designers, engineers, etc.

- Project stakeholders: all individuals and/or organizations who participate in a project, an influence o are incluenced by the project work and results. Stake holders can influence the project work even if they are not involved.
  - Example: new underground line is being built near you house. You are a stake holder even if you are not involved in the actual contruction.

- Program management: the coordinated management of multiple projects, which have similarities - similar goals, similar resources, etc. This tipically involves multiple PMs working with each other in multiple projects.



- Portfolio management: The coordinated management of multiple programs and projects.



  - For example in a pharmaceutical company there might be several drugs being developed at the same time, this will be handled as diffrent project and inside a portfolio.

Chapter 2

The project phases

2.1 How does a project start and evolve? What a project's main stages?

- It has 5 phases:

  1. Initiation: the foundation of a project, a justification of why it is neededed, this will include a suggested budget and time frame.
  2. Planning: create a plan in which the PM needs to estimate the work, risks, resources. Find out the activities, scheduling and timing, budget, project roles. All of the research done in this phase will be included in a document named project management plan. Planning = strategy for winning.
  3. Execution: the actual activities start, all the members of the team are assigned work.
  4. Monitoring and control: this does not start after execution but rather is done at the same time as the execution phase and answers the following question:
     - Progress done? how well are we doing with respect to the time frame? the budget used will be enough? any delays or gaps?
  5. Closure phase: the final tasks, handing the project to the person who will use it, the person recieving it must sign off and confirm that it is complying with the terms. Them the PMs, stake holders and team sit down and share the lessons learned in the just finished project, feedback is collected and rest until the next project begins.

Chapter 3

The initiation phase

3.1 What is the initiation phase about? Define project goals

- Starts with an idea or need, the person who comes up or identify the need is the project sponsor.
- The responsability of this person are the following:
  - Feasibility?
  - Do we have the resources to complete it?
  - Will the project deliver the expected benefits?
  - Appointing a project manager.

3.1.1 Goal

- Defining a goal, remember the definition of a project, that of a temporary initiative that is agreed, planned, and executed to achieve a specific goal.
- The goal must be worthy of being granted a project.

3.2 What is involved in a business case?

- A business case is a document intended to convince management to invest company resources in this project and not in somebody elses.
- It must outline the benefits and the cost (including taxes, inflation, etcetera), the purpose of the project. This is a document that the business leader and project sponsor prepares.
- A scope is defined, a scope includes all activities and questions that belong to the project. For example in building a showroom project you might ask wether or not taking pictures of the not yet completed showroom is part of the project, this is a gray area.
- A gray area is questions and situation which arise during the implementation of a project with respect to what is actually part of a project.
- The project team includes production, IT, R&D,
- A project manager must spot and say in advance and in the initiation phase whether or not something is realistic. If the PM finds a project not feasable the PM must decline to participate in the project.

3.3 Who performs the feasability study and what does it involve?

- A business case details all the finances with respect to the project.
- A scope outlines what yout project will involve.
- A feasability study analyses the goals, scope and the resources to see if they are feasable.

3.3.1 Feasability study

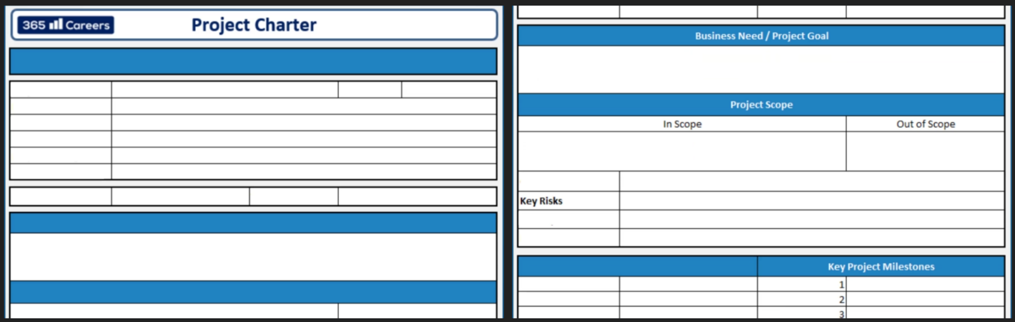
- The project sponsor and PM should be efficient and analyze this quickly to check generally if the budget is feasable, if the organization posses the needed expertise, do partner companies need to be involved, etcetera.

3.4 What goes into risk assesment? What are the expectations?

- Assuming the project is feasable, there are always black swans in the world and bad luck can strike. What are the impacts and risk of this project? This is answered in the risk assesment document.
- A risk assesment also checks what an go wrong in order to plan in advance what can we do to mitigate those risks.
- This document is also used to outline the expectations, outline who is accountable.
- We must spend time with stake holders in order to manage expectations and listen for potential gaps between what is actually being proposed and how to the project is going to look once terminated.

3.5 How to create a project charter?

- Assuming the project is still viable despite the risk assesment and feasability.
- A project charter is a high level document that is quick and easy to understand. An example:



- You can also include the business cases in this document. The important thing is to make this document as easy to read as we can, any one can read it, find what they need and move on. If what they need is more details, the charter will show them where to find the details they need.

3.5.1 Recap

- Initiation phase is the foundation for the entire project.
- The PM can be appointed at any time during the initiation phase so the steps in the initiation phase do not need to be executed in cronological order.

3.6 Case study 1

Outlines the story of an insecure BA that transformed herself into a PM and realized the following:

- No expertise in anything but PM is required to be a PM in a project.
- A PM needs to adapt to new environments quickly.
- A PM is always learning.

Chapter 4

The plannng phase

4.1 What is planning? Why is it important? What happens if it is not done correctly?

- Planning: process of analyzing, evaluate, decide and organize activities in advance.
- Three fundamental steps: define the goal, evaluate options, choose and confirm the best option to achieve the goal.
- Planning is on 2 levels: strategic and tactical.
- Failing to plan properly can end in terrible results.

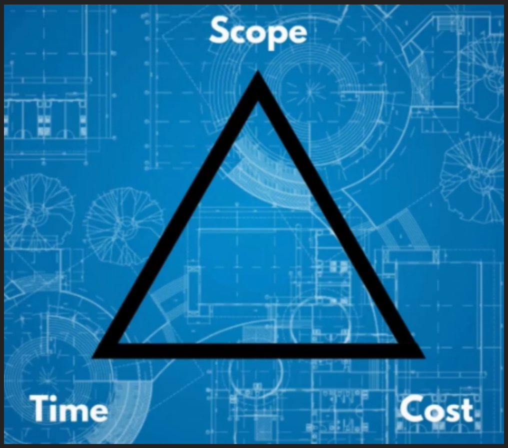
4.2 Why is planning so critical in PM?

- Planning is the fundamental work of the project manager.
- It is when the PM demontrates their expertise.
- You cand devide planning in 2:
  - Things the PM does not know.
  - Things the PM does know and can't control. Here risk assesment, estimations and forecasts need to be covered, this is done using planning.
- Good planning gives great results: optimize work, ensures same expectations, avoids costly errors. The better planning the more chances of success the project has.
- People also work much better with direction, when they understand what to do, how to do it and when they must do it.
- Planning is your strategy to succeed. Despite this planning is often criminally underestimated. People think that the sooner they execute (without planning) they better but in reality the more effort is expended in the planning phase the less effort is needed in during the execution phase. This is the 80/20 rule. More planning means less replanning and delays during execution.

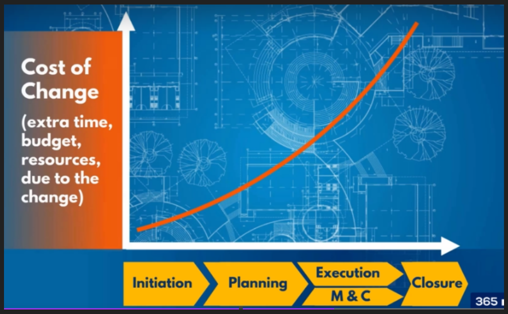
4.3 What is the cost of change in projects?

- Each change to the original project outlined in the project charter impacts cost and time, often both. The aditional budget, time, resources, and coordination efforts are the "cost.

- Remember the triangle of contraints:



- When the change is made also impacts, since it is cheaper if the change is thought out during the initial phase of the project than if the project is near the end.
- The more time passes during execution of the project the higher cost of change it has:



- A PM must take into account potential changes the project might need in advance and that is why they need to be master planners.

4.4 What to do before you start?

- There are some considerations the PM needs to take into account during planning:
  1. The PM must asses their own knowledge of the required work. (Is it enough to plan?)
  2. The PM must know all the project stakeholders.
  3. The PM must recall lessons learned. (Has this project been done before? If so what lessons can we get from the previous projects?)
  4. What about knowledge gaps? (a good project manager is not someone who knows everything, but someone who can identify their own shortcomings and find the right people to compensate for them.)
- A PM must determine how much time they need to put all the pieces of a project together. Also decide with who they will keep in contact with. And also reafirm the expectations to stakeholders.
- In meetings with stakeholders they must prepare with an agenda, a goal, and a comprehensive list of topics to cover.
- PM need to plan for everything.

4.5 Project management insights

- Changes are frequent during planning.
  - New information can arise at any time.
- Planning simple tasks:
  - A simple task must be planned too.
  - A meeting needs to be planned out by determining the participants, day time, room for the meeting, flipchart, and sending out the invitations.
- Tasks with unplanned time durations:
  - Some tasks you cannot estimate, especially if they depend on other people to come to fruition. These tasks can become problematic and can delay the project if not managed correctly.
  - Example of hiring an engineer, you write to HR, after a day they respond they need more details, you reach out to IT in order to get the details, 2 days pass and you have the details, but the details need to be verified by the head of IT that is in vacations and thus what could have easily been planned to take 2 days become 8+ days.

4.6 Scope planning

- At this stage the PM should have information enough to start planning. The planning has its own structure.
- The first thing a PM needs to do is to define the project scope.

4.6.1 Setting up a scope

- Establish the questions that need to be answered.
- In the lamborarri showroom example we need to define how many floors will it have, who will be in charge of what, etcetera.
- Remember scope means what exactly does the project team need to do.

- Once the goal has been established we can then define deliverables.
  - A deliverable must have all the requirements clearly established.
  - An example of deliverables is the following:

- For example lets say that in the land in which we plan to build the show room there is already a building, we need to factor this into our scope since we cannot do anything related to the actual project until we demolish the building.
- Product scope: the tasks that need to be done to produce the product.
- Project scope: all that you need to do in order to achieve the project goal. This includes the product scope.
- A scope needs to be described in detail, a scope definition needs to include what is not in the scope also.
- Take into account that we are not looking for cent precision in planning, if you forgot some trivial thing then that can be added later on with no major incident but if you forgot something big such as a million dollars worth of details then that will be a problem.
- In the showroom project this needs to be agreed between the construction company and the city.

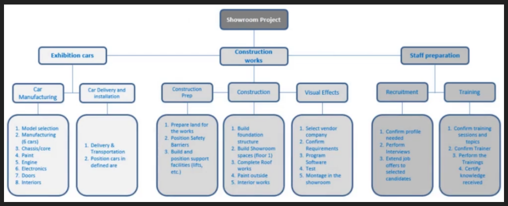
4.6.2 Scope steps

- Assesing information on the project scope.
- Gathering details on project requirement and expectation.
- Using expertise to cover grey areas.
- Documenting everything. (If something arises that need to be checked then it must be available.)

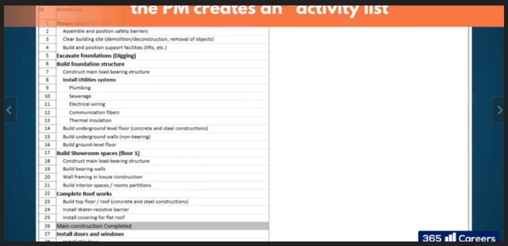
4.7 Scope part 2

- Recap:
  - We analyze the available information.
  - We gather detailed requirements and expectations.
  - We document the scope.
- Now for a low level look at scope planning:
  - The PM analyses high-level information gained in the initiation stage from the project charter, discussions with the sponsor and clients, as well as stakeholders.
  - Then gather detailed requirements and expectations, such as the sponsors and managers, the sponsor and senior manager are a reliable source of information on scope if the PM asks the right questions.
  - The more information the PM gathers the better.
  - Then with all this information the PM must document all this in a document called a scope statement document. This document is looked at as a sort of contract, between the PM and stakeholders, it outlines what the PM has commited and has not commited to do.
- The scope statement document is not user friendly.

- This is why the scope statement needs to be reformated and split into deliverables and then each deliverable must be assigned to a specific person.



- For an even more user-friendly experience the PM creates an activity list. Which details the task, person, activity, etcetera. This to make it absolutely clear what will be done, by who and by which date. This document can be refered to by anyone.



- Note that there are three documents involved here that contain the same information in different formats: